Quiz 3

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Contents

Please note the quiz is based on the following topics:

- Conjugation.
- Normal Subgroups.

Problem 1:
In the group D_4, let $r = (1234)$ be a clockwise rotation, and let $s = (12)(34)$ be a reflection. Which of the following elements are equal to sr?

(a) rs.

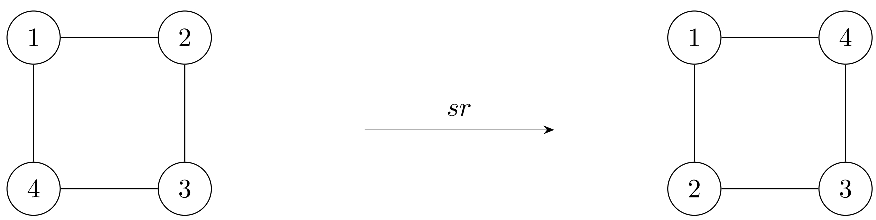
Solution:

rs does not equal sr.

For this and the following questions, we will use the fact that:

$$\begin{aligned} sr &= (12)(34)(1234) \\ &= (24) \end{aligned}$$

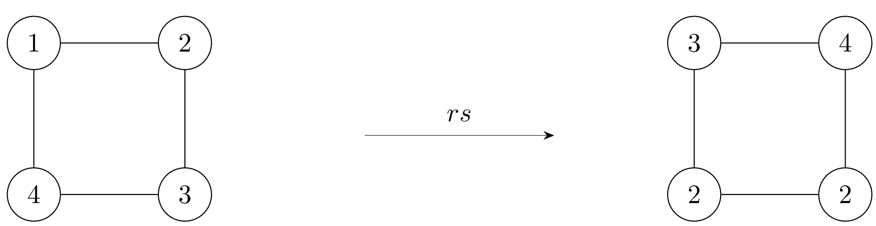
Which graphically looks like the following adjacency-preserving permutation of the vertices of a square:



to determine whether or not the given permutations are equivalent to sr. In this case, since:

$$\begin{aligned} rs &= (1234)(12)(34) \\ &= (13) \end{aligned}$$

Which can be graphically represented by the transformation:



It is clear that rs and sr are different permutations, and thus, are unequal.

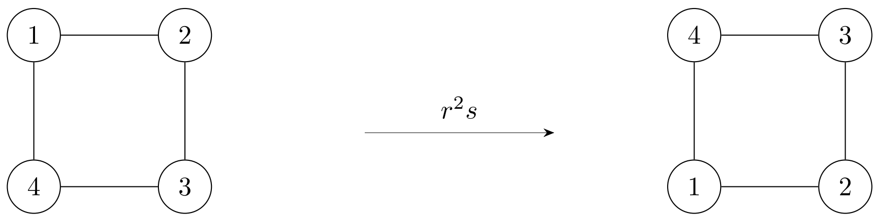
(b) r^2s.

Solution:

r^2s doesn't equal sr as well. To see why, we can leverage the fact that $rs = (13)$ (which we determined in part **(a)**) to conclude that:

$$
\begin{aligned}
r^2 s &= r(rs) \\
&= (1234)(13) \\
&= (14)(23)
\end{aligned}
$$

Which visually looks like this:



And since this is clearly a different permutation than sr, $r^2 s \neq sr$.

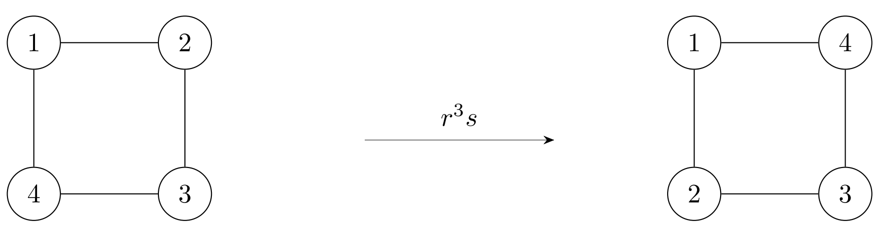
(c) $r^3 s$.

Solution:

$r^3 s$ actually does equal sr!
Proceeding as we did before, since:

$$
\begin{aligned}
r^3 s &= r(r^2 s) \\
&= (1234)(14)(23) \\
&= (24)
\end{aligned}
$$

Which looks like:



We can see that $r^3 s$ is indeed the same permutation as sr, and so $r^3 s = sr$.

(d) $r^4 s$.

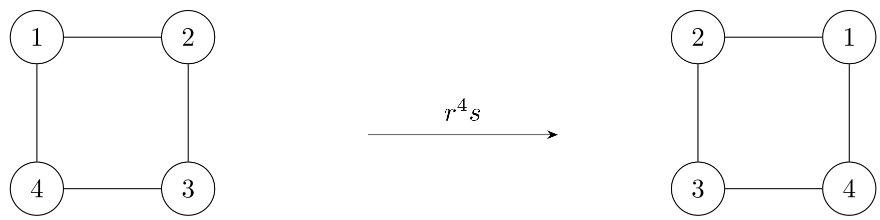
Solution:

$r^4 s$ doesn't equal sr.

This is because:

$$\begin{aligned} r^4s &= r(r^3s) \\ &= (1234)(24) \\ &= (12)(34) \end{aligned}$$

Or, graphically:



Thus, r^4s and sr are clearly different permutations and so $r^4s \neq sr$.

Problem 2:
Let G be a group, and let $a \in G$. What notations represent the conjugacy class of a?

Solution:

a^G and $[a]$ are both ways of denoting the conjugacy class of a. Note that a^G is perhaps a little more descriptive than $[a]$, since it explicitly states which group it is a conjugacy class of.

Problem 3:
Let G be a group and let $a, b \in G$. What is the definition of a^b?

Solution:

a^b is defined by:

$$a^b = b^{-1}ab$$

Problem 4:
If $\sigma = (153)(24)$ and $g = (5314)$, then what does σ^g equal?

Solution:

By the definition of σ^g given as the solution to the problem just above, we know that:

$$\begin{aligned}
\sigma^g &= g^{-1}\sigma g && \text{(By definition of conjugation)} \\
&= (5314)^{-1}(153)(24)(5314) && \text{(Substituting)}
\end{aligned}$$

So, to compute σ^g, we'll need to find g^{-1}! This is thankfully not very difficult to do. Although I won't prove it (:P), it is generally true that the inverse of any permutation of the form $(abc \ldots z)$ is just $(zxy \ldots a)$, and so $g^{-1} = (4135)$.

So carrying out our earlier computation, we have that:

$$\begin{aligned}
\sigma^g &= (5314)^{-1}(153)(24)(5314) && \text{(From earlier equation)} \\
&= (4135)(153)(24)(5314) && \text{(Since } (5314)^{-1} = (4135)\text{)} \\
&= (4135)(12435) && \text{(Simplifying)} \\
&= (12)(345) && \text{(Simplifying)}
\end{aligned}$$

Thus, $\sigma^g = (12)(345)$.

Problem 5:
Find all elements $g \in S_3$ such that $g^{-1}(123)g = (132)$.

Solution:

We know that $S_3 = \{\, (), (12), (23), (13), (123), (132) \,\}$ so, together with the quick and easy method for finding the inverse of a permutation I mentioned in the solution to the problem above, we can brute force all possible $g \in S_3$ to find that:

(I) When $g = ()$:

$$\begin{aligned} g^{-1}(123)g &= ()(123)() \\ &= ()(123) \\ &= (123) \end{aligned}$$

(II) When $g = (12)$:

$$\begin{aligned} g^{-1}(123)g &= (21)(123)(12) \\ &= (21)(13) \\ &= \boxed{(132)} \end{aligned}$$

(III) When $g = (23)$:

$$\begin{aligned} g^{-1}(123)g &= (32)(123)(23) \\ &= (32)(12) \\ &= \boxed{(132)} \end{aligned}$$

(IV) When $g = (13)$:

$$\begin{aligned} g^{-1}(123)g &= (31)(123)(13) \\ &= (31)(23) \\ &= \boxed{(132)} \end{aligned}$$

(V) When $g = (123)$:

$$\begin{aligned} g^{-1}(123)g &= (321)(123)(123) \\ &= (321)(132) \\ &= (123) \end{aligned}$$

(VI) When $g = (132)$:

$$\begin{aligned} g^{-1}(123)g &= (231)(123)(132) \\ &= (231)() \\ &= (231) \\ &= (123) \end{aligned}$$

Thus, the set of all g in S_3 satisfying $g^{-1}(123)g = (132)$ is $\{\,(12),(23),(13)\,\}$.

Problem 6:
How many elements do the following conjugacy classes contain?

(a) $[(123)]$ in S_4.

Solution:

$[(123)]$ in S^4 has 8 elements.

By Theorem 5.3, we know that (123) is conjugate to another permutation, $\tau \in S_4$, if and only if (123) and τ have the same cycle type.

The cycle type of (123) in S_4 is $(3, 1)$ and so all permutations in S_4 conjugate to (123) must likewise have a cycle type of $(3, 1)$. Therefore, all permutations in S_4 conjugate to (123) are expressible as (abc), where $a, b, c \in \{1, 2, 3, 4\}$ and a, b, and c are all pairwise distinct.

Clearly then, there are $4! = 24$ possible such permutations in S_4, since:

- There are 4 possible choices for a.
- 3 choices for b so as to not conflict with a.
- 2 for c to avoid conflicting with both a and b.
- Only one choice as to which element the permutation leave fixed, after a, b, and c have all been chosen.

However, recall that the permutations (abc), (bca), and (cab) are all equivalent, meaning 24 over counts the number of distinct such permutations in S_4 by a factor of 3. Thus, there are actually $\frac{24}{3} = 8$ permutations with a cycle type of $(3, 1)$ in S_4.

Thus, since there are exactly 8 permutations in S_4 with the same cycle type as (123), there are also exactly 8 elements in the conjugacy class of (123).

(b) $[(123)]$ in A_4.

Solution:

(c) $[5]$ in $\mathbb{Z}_6^\times$.

Solution:

$[5]$ in $\mathbb{Z}_6^\times$ has 1 element.

$\mathbb{Z}_6^\times$ is a relatively small group, so we will find the size of $[5]$ in $\mathbb{Z}_6^\times$ by brute force. By Proposition 1.7, we know that because:

$$\begin{array}{cc} \gcd(0,6) = 6 & \gcd(3,6) = 3 \\ \boxed{\gcd(1,6) = 1} & \gcd(4,6) = 2 \\ \gcd(2,6) = 2 & \boxed{\gcd(5,6) = 1} \end{array}$$

1 and 5 are the only units modulo 6. Thus, $\mathbb{Z}_6^\times = \{1, 5\}$. Now, since:

$$\begin{aligned}
5^1 &\equiv (1)^{-1}(5)(1) \pmod 6\\
&\equiv (1)(5)(1) \pmod 6\\
&\equiv (1)(5) \pmod 6\\
&\equiv 5 \pmod 6\\
&\equiv 5
\end{aligned}$$

and:

$$\begin{aligned}
5^5 &\equiv (5)^{-1}(5)(5) \pmod 6\\
&\equiv (5)(5)(5) \pmod 6\\
&\equiv (5)(25) \pmod 6\\
&\equiv 125 \pmod 6\\
&\equiv 5
\end{aligned}$$

We therefore have that the conjugacy class of 5 in $\mathbb{Z}_6^\times$ is just the set containing 5. Thus, since $[5] = \{ 5 \}$, $[5]$ contains only one element.

Problem 7:
How many conjugacy classes are there in S_5.

Solution:

There are 7 conjugacy classes in S_5.

By Theorem 5.3 and the fact that conjugacy classes form equivalence partitions over their domain, we know there must be as many conjugacy classes in S_5 as there are distinct cycle types in S_5.

Since there are 7 distinct cycle types in S_5, given by the set $\{\,(1,1,1,1,1),(2,1,1,1),(2,2,1),(3,1,1),(3,2),(4,1),(5)\,\}$, (derived from the integer partitions of 5), there must therefore be 7 conjugacy classes in S_5, as argued earlier.

Problem 8:
Which of the following subgroups are normal subgroups of $G = GL_2(\mathbb{Z}_2)$?

(a) $H = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \right\}$

Solution:

H is a normal subgroup of G.

To see why, recall that for H to be a normal subgroup of G, it must be closed under conjugation, meaning that for all $g \in G$ and $h \in H$, $h^g \in H$. Since I couldn't really think of a better way, our approach will be to compute h^g for all possible h and g, but cleverly take shortcuts where possible to cut down on the number of required calculations.

For one, since $h^g = g^{-1}hg$, it'd be convenient to have an easy formula for g^{-1}, given g, that we can use to plug and chug with. It is for that reason that I will quickly sidetrack to prove that for any $g \in G$ of the form:

$$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$g^{-1} \in G$ is given by:

$$g^{-1} = \begin{pmatrix} d & b \\ c & a \end{pmatrix}$$

To understand how this is the case, all that's required is to understand two things:

1. For any $x \in \mathbb{Z}_2, 2x = 0$.
   - This is because if $x = 0$, then $2x = (2)(0) \equiv 0$, and if $x = 1$, then $(2)(1) = 2 = 0 \pmod 2$.
2. $ad + bc = 1$.
   - This is because exactly one of ad or bc can be 1, and the other must be 0.
     - Since $\det(g) = ad - bc \neq 0$, by definition of g belonging to $GL_2(\mathbb{Z}_2)$.

And so, we have that:

$$\boxed{\begin{pmatrix} d & b \\ c & a \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix}} = \begin{pmatrix} ad+cb & db+bd \\ ca+ac & cb+ad \end{pmatrix} \quad \text{(Simplifying)}$$

$$= \begin{pmatrix} ad+bc & 2bd \\ 2ac & ad+bc \end{pmatrix} \quad \text{(Simplifying)}$$

$$= \begin{pmatrix} ad+bc & 0 \\ 0 & ad+bc \end{pmatrix} \quad \text{(Since } 2x = 0 \text{ for all } x \in \mathbb{Z}_2\text{)}$$

$$= \boxed{\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}} \quad \text{(Since } ad+bc = 1\text{)}$$

$$= \begin{pmatrix} bc+ad & 0 \\ 0 & bc+ad \end{pmatrix} \quad \text{(Since } bc+ad = 1\text{)}$$

$$= \begin{pmatrix} bc+ad & 2ab \\ 2cd & bc+ad \end{pmatrix} \quad \text{(Since } 2x = 0 \text{ for all } x \in \mathbb{Z}_2\text{)}$$

$$= \boxed{\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} d & b \\ c & a \end{pmatrix}}$$

And thus, $g^{-1} = \begin{pmatrix} d & b \\ c & a \end{pmatrix}$. So now that we know how to compute $g^{-1}hg$, we need only consider 3 possible cases, recalling that we need only check that h is closed under conjuigation by $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$, and $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, since all other matrices in $GL_2(\mathbb{Z}_2)$ belong to H, which is given to be a group.

(I) $h = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

If $h = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, then we wouldn't even need to consider the different choices of g to see that:

$$h^g = g^{-1}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} g \quad \text{(By definition of conjugation)}$$

$$= g^{-1}g \quad \text{(By definition of the identity)}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{(By definition of inverses)}$$

And because $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \in H, h^g \in H$ in this case. (Note that we also could've determined this from the fact the conjugacy class of the identity element is always the set containing just itself).

(II) $h = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$.

Here, we'll need to consider each possible g individually.

(a) $g = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

If $g = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, then $g^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, and so we can compute:

$$\begin{aligned}
h^g &= g^{-1}hg && \text{(By definition of conjugation)} \\
&= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} && \text{(Substituting)} \\
&= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} && \text{(Simplifying)} \\
&= \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} && \text{(Simplifying)}
\end{aligned}$$

And since $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \in H, h^g \in H$ in this case.

(b) $g = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$.

If $g = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$, then $g^{-1} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$, and so we can compute:

$$\begin{aligned}
h^g &= g^{-1}hg && \text{(By definition of conjugation)} \\
&= \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} && \text{(Substituting)} \\
&= \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} && \text{(Simplifying)} \\
&= \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} && \text{(Simplifying)}
\end{aligned}$$

And since $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \in H, h^g \in H$ in this case as well.

(c) $g = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

If $g = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, then $g^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, and so we can compute:

$$\begin{aligned}
h^g &= g^{-1}hg && \text{(By definition of conjugation)} \\
&= \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} && \text{(Substituting)} \\
&= \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} && \text{(Simplifying)} \\
&= \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} && \text{(Simplifying)}
\end{aligned}$$

And since $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \in H, h^g \in H$ once again.

Thus, since $h^g \in H$ in cases **(a)**, **(b)**, and **(c)**, we have that $h = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ in H is closed under conjugation.

(III) $h = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$.

We'll once again have to consider each case of g separately.

(a) $g = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

When $g = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, then $g^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, and so:

$$
\begin{aligned}
h^g &= g^{-1}hg && \text{(By definition of conjugation)}\\
&= \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix}1 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} && \text{(Substituting)}\\
&= \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix}1 & 1\\ 0 & 1\end{pmatrix} && \text{(Simplifying)}\\
&= \begin{pmatrix}0 & 1\\ 1 & 1\end{pmatrix} && \text{(Simplifying)}
\end{aligned}
$$

And because $\begin{pmatrix}0 & 1\\ 1 & 1\end{pmatrix} \in H$, then $h^g \in H$ in this case.

(b) $g = \begin{pmatrix}1 & 0\\ 1 & 1\end{pmatrix}$.

Assuming $g = \begin{pmatrix}1 & 0\\ 1 & 1\end{pmatrix}$, then $g^{-1} = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}$ and:

$$
\begin{aligned}
h^g &= g^{-1}hg && \text{(By definition of conjugation)}\\
&= \begin{pmatrix}1 & 0\\ 1 & 1\end{pmatrix}\begin{pmatrix}1 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix}1 & 0\\ 1 & 1\end{pmatrix} && \text{(Substituting)}\\
&= \begin{pmatrix}1 & 0\\ 1 & 1\end{pmatrix}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} && \text{(Simplifying)}\\
&= \begin{pmatrix}1 & 1\\ 1 & 0\end{pmatrix} && \text{(Simplifying)}
\end{aligned}
$$

And since $\begin{pmatrix}1 & 1\\ 1 & 0\end{pmatrix} \in H$, h^g belongs to H in this case as well.

(c) $g = \begin{pmatrix}1 & 1\\ 0 & 1\end{pmatrix}$.

Finally, if $g = \begin{pmatrix}1 & 1\\ 0 & 1\end{pmatrix}$, then $g^{-1} = \begin{pmatrix}1 & 1\\ 0 & 1\end{pmatrix}$ and thus:

$$
\begin{aligned}
h^g &= g^{-1}hg && \text{(By definition of conjugation)} \\
&= \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} && \text{(Substituting)} \\
&= \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} && \text{(Simplifying)} \\
&= \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} && \text{(Simplifying)}
\end{aligned}
$$

And since $\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \in H$, h^g belongs to H in this case as well.

Therefore, since h^g belongs to H in cases **(a)**, **(b)**, and **(c)**, we can therefore conclude that $h = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ in H is closed under conjugation.

Thus, since cases **(I)**, **(II)**, and **(III)** all show that the elements of H are closed under conjugation, H is closed under conjugation as well and is thus a normal subgroup of G. Phew!

(b) $K = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right\}$.

Solution:

K is not a normal subgroup of G. To see why, all we need is to observe that when $k = \begin{pmatrix} 0 & 1 & 1 & 0 \end{pmatrix}$ and $g = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$:

$$
\begin{aligned}
k^g &= g^{-1}kg && \text{(By definition of conjugation)} \\
&= \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} && \text{(Substituting)} \\
&= \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} && \text{(Simplifying)} \\
&= \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}
\end{aligned}
$$

which doesn't belong to K, despite the fact that $k \in K$ and $g \in G$! Thus, K cannot be closed under conjugacy and thus doesn't constitute a normal subgroup of G.

Problem 9:
Consider the following proposition:

Proposition: Conjugation satisfies the following:

(**a**) Let $a, g, h \in G$. Then $a^{gh} = (a^g)^h$.

(**b**) Let $a, b, g \in G$. Then $(ab)^g = a^g b^g$.

Now, suppose that the definition of a^g was changed to $a^g = gag^{-1}$. How would claim **(a)** and **(b)** made by the proposition change, if at all?

Solution:

We'll consider the impact that this new definition of conjugacy would have on claims **(a)** and **(b)** separately.

(I) How would claim **(a)** change?

Claim **(a)** would change to become $a^{gh} = (a^h)^g$.

Using this new definition of conjugacy, we can see that for arbitrary $a, g, h \in G$:

$$\begin{aligned}
a^{gh} &= (gh)a(gh)^{-1} && \text{(By our new definition of conjugation)} \\
&= (gh)a(h^{-1}g^{-1}) && \text{(By exponent laws)} \\
&= g(hah^{-1})g^{-1} && \text{(By associativity)} \\
&= g(a^h)g^{-1} && \text{(By our new definition of conjugation)} \\
&= (a^h)^g && \text{(By our new definition of conjugation)}
\end{aligned}$$

And thus, $a^{gh} = (a^h)^g$ when conjugacy is defined this way.

(II) How would claim **(b)** change?

Claim **(b)** wouldn't change at all!

Using this new definition of conjugacy, we can see that for arbitrary $a, b, g \in G$:

$$\begin{aligned}
(ab)^g &= g(ab)g^{-1} && \text{(By our new definition of conjugation)} \\
&= g(a(1)b)g^{-1} && \text{(By definition of identity)} \\
&= g(a(g^{-1}g)b)g^{-1} && \text{(By definition of inverse)} \\
&= (gag^{-1})(gbg^{-1}) && \text{(By associativity)} \\
&= a^g b^g && \text{(By our new definition of conjugation)}
\end{aligned}$$

Thus, even with the definition of conjugacy changes, $(ab)^g$ still equals $a^g b^g$.

Problem 10:
Which of the following statements are true?

(a) In an abelian group, every subgroup is normal.

Solution:

This statement is true.

Let G be any abelian group. To prove the statement true, we'll need to show that every subgroup of G, $H \leq G$, is a normal subgroup. So, let $H \leq G$ be an arbitrary subgroup of G, and let $h \in H$ and $g \in G$ be arbitrary elements of H and G respectively.

By definition of normalcy, H is a normal subgroup of G if $h^g = g^{-1}hg \in H$. Recall, however, that G is abelian, and so for arbitrary elements x and y, $xy = yx$. Thus, we know that:

$$\begin{aligned} h^g &= g^{-1}hg && \text{(By definition of conjugation)} \\ &= g^{-1}gh && \text{(By commutativity, since } G \text{ is abelian)} \\ &= (1)h && \text{(By definition of inverses)} \\ &= h && \text{(By definition of identity)} \end{aligned}$$

And since $h \in H$ by assumption, we have that $h^g \in H$. Thus H constitutes a normal subgroup of G. Since H was arbitrary, we can thereby conclude that every subgroup of an abelian group G must be normal.

(b) Every group has at least two normal subgroups.

Solution:

(c) The intersection of two normal subgroups is also a normal subgroup.

Solution:

This statement, once again, is true!

Let G be an arbitrary group, and let $H \trianglelefteq G$ and $K \trianglelefteq G$ be any two normal subgroups of G. To prove the statement correct, we need to show that $H \cap K$ is itself always a normal subgroup of G. To do that, we need to show that $x^g \in (H \cap K)$ for all $x \in (H \cap K)$ and $g \in G$.

So, let $x \in (H \cap K)$ and $g \in G$ be arbitrary elements of their respective groups. By definition of set intersection, we know that $x \in H$ and $x \in K$. Since H and K were assumed to be normal subgroups of G, we therefore know that $x^g \in H$ <u>and</u> that $x^g \in K$, since both H and K must both be closed under conjugation. However, since $x^g \in H$ and $x^g \in K$, we have that $x^g \in (H \cap K)$.

Thus, since $x \in (H \cap K)$ and $g \in G$ implies $x^g \in (H \cap K)$, $H \cap K$ is closed under conjugation, making $H \cap K$ a normal subgroup of G. Because H, K, and G were all arbitrary, this altogether proves the statement.

(d) If A is a normal subgroup of B, and B is a normal subgroup of C, then A is a normal subgroup of C.

Solution:

This statement is false.

We will prove this statement false by use of a neat little counter example. Consider the groups S_3, A_3, and $(123)^{A_3}$. To prove that the statement doesn't hold, we'll show that although $(123)^{A_3} \trianglelefteq A_3$ and $A_3 \trianglelefteq S_3$, $(123)^{A_3}$ is not a normal subgroup of S_3.

Our first order of business will be to show that $(123)^{A_3} \trianglelefteq A_3$ and $A_3 \trianglelefteq S_3$.

(I) Prove $(123)^{A_3} \trianglelefteq A_3$.

This statement holds definitionally. Since $(123)^{A_3}$ is the set of all permutations in A_3 that (123) is conjugate to, and since conjugacy constitutes and equivalence relation over its domain, $(123)^{A_3}$ is necessarily closed under conjugation, making $(123)^{A_3}$ a normal subgroup of A_3.

(II) Prove $A_3 \trianglelefteq S_3$.

Here, we'll try a clever approach to show that A_3 is closed under conjugation by relying on the very definition of A_3 itself.

Let $\sigma \in A_3$ and $\tau \in S_3$ be arbitrary permutations in their respective groups, and let P_σ and P_τ be their associated permutation matrices. By definition of A_3, we know that $\det(P_\sigma) = 1$ since A_3 is the set of all even permutations in S_3.

Because $\det(AB) = \det(A)\det(B)$ for arbitrary matrices in $M_n(\mathbb{R})$, and **Proposition 3.13 (a)** implies that $\det(P_{\sigma^{-1}}) = \frac{1}{\det(P_\sigma)}$ since:

$$
\begin{aligned}
\det(P_{\sigma^{-1}})\det(P_\sigma) &= \det(P_{\sigma^{-1}}P_\sigma) && \text{(By aforementioned property of the determinant)} \\
&= \det(P_{\sigma\sigma^{-1}}) && \text{(By \textbf{Proposition 3.13 (a)})} \\
&= \det(P_{()}) && \text{(By definition of inverse)} \\
&= \det(I_3) && \text{(By definition of permutation matrices)} \\
&= 1 && \text{(By definition of the identity)}
\end{aligned}
$$

We can thus conclude that:

$$
\begin{aligned}
\det(P_\sigma^\tau) &= \det(P_{\tau^{-1}\sigma\tau}) && \text{(By definition of conjugation)} \\
&= \det(P_{\tau^{-1}}P_\sigma P_\tau) && \text{(By \textbf{Proposition 3.13 (a)})} \\
&= \det(P_{\tau^{-1}})\det(P_\sigma)\det(P_\tau) && \text{(By aforementioned property of determinants)} \\
&= \frac{1}{\det(P_\tau)}\det(P_\sigma)\det(P_\tau) && \text{(Since } P_{\sigma^{-1}} = \frac{1}{P_\sigma}\text{)} \\
&= \det(P_\sigma) && \text{(Simplifying)} \\
&= 1 && \text{(By assumption } \sigma \in A_3\text{)}
\end{aligned}
$$

And so, since σ^τ clearly belongs to S_3, and its associated permutation matrix P_{σ^τ} has a determinant of 1, we know that σ^τ is an even permutation and so belongs to A_3, by definition.

Because $\sigma \in A_3$ and $\tau \in S_3$ were arbitrary, we can therefore conclude that A_3 is closed under conjugation and is thus a normal subgroup of S_3, as required.

Thus, considerations **(I)** and **(II)** together tell us that $(123)^{A_3} \trianglelefteq A_3$ and $A_3 \trianglelefteq S_3$. But is $(123)^{A_3}$ a normal subgroup of S_3? Well, since A_3 is an abelian group, we can appeal to part **(a)** of this problem to conclude that $(123)^{A_3} = \{\, (123) \,\}$. However, we also know that $(12) \in S_3$ satisfies:

$$\begin{aligned}(123)^{(12)} &= (21)(123)(12)\\ &= (21)(13)\\ &= (123)\end{aligned}$$

And yet, (132) doesn't belong to $(123)^{A_3}$! Thus, $(123)^{A_3}$ isn't closed under conjugation and so cannot be a normal subgroup of S_3.

Therefore, since $(123)^{A_3} \trianglelefteq A_3$ and $A_3 \trianglelefteq S_3$, but $(123)^{A_3} \not\trianglelefteq S_3$, we found a counter example to the claim that normalcy is transitive.

(e) G is always a normal subgroup of itself.

Solution:

This statement is true, and is fairly easy to prove!

To show G is always its own subgroup, we need to show that $a^b \in G$ for all $a, b \in G$. Since G is a group, we know it must be closed under its group operation. Thus, there must be some $c \in G$ such that $ab = c$. Furthermore, G being a group also guarantees to us that G is closed under inversion as well. Thus, b^{-1} likewise belongs to G, implying there must exist some $d \in G$ to satisfy $d = b^{-1}c$.

Therefore, because:

$$\begin{aligned}a^b &= b^{-1}ab && \text{(By definition of conjugation)}\\ &= b^{-1}c && \text{(By definition of } c\text{)}\\ &= d && \text{(By definition of } d\text{)}\end{aligned}$$

and $d \in G$ by assumption, then a^b likewise belongs to G! Thus, G is closed under conjugation and is thereby a normal subgroup of itself.

(f) Every group is partitioned into its conjugacy classes.

Solution:

This statement is true!

Recall that a group G is said to be partitioned by a collection of sets $\{\, P_1, \ldots, P_n \,\}$ if the following two criteria are satisfied:

1. For all $g \in G$, there exists some P_i such that $x \in P_i$.
2. $P_i \cap P_j = \emptyset$ for all $i \neq j$.

So, to prove that a group G is partitioned by its conjugacy classes, we need to show that:

(I) Every element of G belongs to some conjugacy class.

This immediately follows as a corollary to **Exercise 5.1**, which asserts thats conjugation constitutes an equivalence relation on any group G.

Since conjugation is an equivalence relation, it must necessarily be reflexive, meaning for arbitrary g in G, g is its own conjugate.

Thus, $g \in [g]$ at the very least, and since $[g]$ is evidently a conjugacy class of G, because $g \in G$ was arbitrary, we therefore have that every element in a group must belong to <u>some</u> conjugacy class.

(II) Every pair of distinct conjugacy classes are pairwise disjoint.

Let P and Q be any two conjugacy classes of G such that $P \neq Q$. To prove that they're pairwise disjoint, we need to show that this implies $P \cap Q = \emptyset$. This thankfully follows quite readily, yet again, from the fact conjugation is an equivalence relation on G.

Let's assume, for the purposes of deriving a contradiction, that $P \cap Q$ is nonempty. There must therefore exist some element $x \in (P \cap Q)$. By definition of set intersection, we thus have that $x \in P$ and $x \in Q$.

Let $p \in P$ and $q \in Q$ be two arbitrary elements of their respective conjugacy classes. Since x and p both belong to P, by definition of conjugacy classes, we know that $p \sim x$. Similarly, since q and x both belong to Q, we also know $x \sim q$.

Since conjugation is an equivalence relation, it must therefore be transitive. Thus, by definition of transitivity, since $p \sim x$ and $x \sim q$, we have that $p \sim q$. But p and q were arbitrary! This means every element in P is conjugate to every element in Q and vice versa (since conjugation is also symmetric, owing to the fact it is an equivalence relation). Hence, by definition of conjugacy classes $P = Q$!

This contradicts our prior assumption that P and Q must be non-equal conjugacy classes of G. Thus, $P \cap Q$ <u>must</u> be empty, meaning any two distinct conjugacy classes satisfy this criterion as well!
Thus, since conjugacy classes of any arbitrary group G always satisfy the criteria needed to partition G (as per the findings of considerations **(I)** and **(II)**), we can therefore conclude that every group must be partitioned by its conjugacy classes.

Problem 11:
In **Chapter 6**, the quaternion group Q_8 serves as a counter example to which conjecture?

Solution:

The conjecture was:

If every subgroup of G is normal, then G must be abelian.

Problem 12:
Complete the following computations in the quaternion group, Q_8.

(a) ij.

Solution:
$ij = k$, since:

$$\begin{aligned}
ij &= ij(1) && \text{(By definition of the identity)}\\
&= ij(-1)^2 && \text{(Expanding)}\\
&= ij(-1)(-1) && \text{(Expanding)}\\
&= ijk^2(-1) && \text{(By fundamental identity of quaternions)}\\
&= ijkk(-1) && \text{(Expanding)}\\
&= (-1)k(-1) && \text{(By fundamental identity of quaternions)}\\
&= k && \text{(Simplifying)}
\end{aligned}$$

(b) ik.

Solution:
$ik = -j$ because:

$$\begin{aligned}
ik &= i(1)k && \text{(By definition of the identity)}\\
&= i(-1)(-1)k && \text{(Expanding)}\\
&= i(ijk)(-1)k && \text{(By fundamental identity of quaternions)}\\
&= -i^2jk^2 && \text{(Simplifying)}\\
&= jk^2 && \text{(By fundamental identity of quaternions)}\\
&= -j && \text{(By fundamental identity of quaternions)}
\end{aligned}$$

(c) ji.

Solution:
$ji = -k$ because:

$$\begin{aligned}
ji &= ji(1) && \text{(By definition of the identity)}\\
&= ji(-1)(-1) && \text{(Expanding)}\\
&= ji(ijk)(-1) && \text{(By fundamental identity of quaternions)}\\
&= -ji^2jk && \text{(Simplifying)}\\
&= j^2k && \text{(By fundamental identity of quaternions)}\\
&= -k && \text{(By fundamental identity of quaternions)}
\end{aligned}$$

(d) i^{-1}.

Solution:
$i^{-1} = -i$, since:

$$
\begin{aligned}
\boxed{(i)(-i)} &= -i^2 && \text{(Simplifying)}\\
&= -(-1) && \text{(By fundamental identity of quaternions)}\\
&= \boxed{1} && \text{(Simplifying)}\\
&= -(-1) && \text{(Expanding)}\\
&= -i^2 && \text{(By fundamental identity of quaternions)}\\
&= \boxed{(-i)(i)} && \text{(Expanding)}
\end{aligned}
$$

(e) $(-j)^{-1}$.

Solution:
$(-j)^{-1} = j$ because:

$$
\begin{aligned}
\boxed{(-j)(j)} &= -j^2 && \text{(Simplifying)}\\
&= -(-1) && \text{(By fundamental identity of quaternions)}\\
&= \boxed{1} && \text{(Simplifying)}\\
&= (-1)(-1) && \text{(Expanding)}\\
&= (j^2)(-1) && \text{(By fundamental identity of quaternions)}\\
&= (j)(j)(-1) && \text{(Expanding)}\\
&= \boxed{(j)(-j)} && \text{(Simplifying)}
\end{aligned}
$$

Problem 13:
How many conjugacy classes does Q_8 have?

Solution: